Stall in Steady RANS Is a Branch-Selection Problem, Not a Convergence Problem

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Abstract

Two standard checks certify a steady RANS solution: grid convergence and iterative convergence. Using a 125-case campaign on the NACA 0012 (structured polar O-grids, $y^+ \approx 0.4$, k–ω SST with a γ–$Re_{\theta t}$ transition block), we show that at trailing-edge stall both checks pass while the answer is still wrong — because the discrete system possesses *multiple steady branches*, and neither refinement nor iteration selects among them. Under grid refinement at $\alpha = 15°$ the upper-surface separation point jumps between three branches ($x/c = 0.53 \to 0.20 \to$ attached), while the same sequence is cleanly convergent at $\alpha = 5°$ and 18°. A doubled-horizon URANS verification ($t^* = 0.81$–2.4, 61 samples) then shows the converged steady state is not an attractor: windowed means drift monotonically away from it, and at $\alpha = 15°$ the steady and time-averaged drags bracket the wind-tunnel measurement. At stall, a single "converged" number is therefore not a well-defined object; branch envelopes and horizon-dependence must be reported instead.

1 Introduction

For attached flows, the certification protocol of computational aerodynamics is settled: refine the grid until a functional changes by a tolerated fraction, iterate until residuals stall, and report with a discretisation-error estimate. This letter asks what the same protocol certifies at *trailing-edge stall onset*, using 125 scripted OpenFOAM runs, three-grid families at three incidences spanning the stall regime, and two time-accurate repeats with a doubled statistical horizon. The findings reduce to two statements: the steady solution set is not single-valued at stall onset (Section 3), and the selected branch is not an attractor of the unsteady system (Section 4).

2 Numerical experiment

Sections are meshed with Gmsh into structured polar O-grids (coarse 16,100, medium 33,740, fine 51,360 cells; $y^+ \approx 0.4$; farfield 20 chords, 50 for verification), closed with k–ω SST and, in a transition block, γ–$Re_{\theta t}$ [1]. Steady cases use `simpleFoam` (4,000 iterations; separation from skin-friction reversal). The URANS subset uses `pimpleFoam` at $Re = 3.6 \times 10^5$ [2] at $\alpha = 12°$ and 15° on the 50-chord medium grid, sampled every 0.005 s. Baseline credibility rests on the Reynolds-matched Sandia free-transition comparison (Figure 1): with the transition closure, pre-stall drag error falls by one third and attached-range lift is within 4–5% of measurement; the bubble-burst stall of the experiment is not reproduced. All 125 cases, meshes, and scripts are released for end-to-end reproduction.

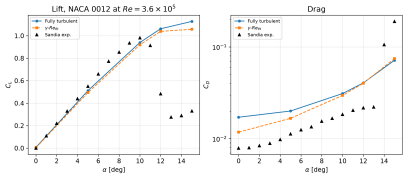


Figure 1: Baseline validation: NACA 0012 at $Re = 3.6 \times 10^5$ with free transition against the Sandia data [2].

3 Result 1: refinement does not select a branch at stall onset

Figure 2a tracks the upper-surface separation point through the refinement sequence (16,100 → 33,740 → 51,360 cells) at three incidences; Table 1 gives the exact forces. At $\alpha = 5°$ the flow is attached on every grid — lift changes by 0.58% and drag by 0.05% from medium to fine. At the post-stall angle $\alpha = 18°$ both grids select the *same* deeply separated branch (drag within 3%). But at the stall-onset angle $\alpha = 15°$ the three grids land on *three different branches* (Table 1) — a 26% lift spread across a monotone refinement sequence. No grid-convergence index can be formed from this family, and none should be: it is not approaching a single limit, it is hopping between basins. The medium-grid production values of the campaign are therefore a *branch sample*, not a converged estimate.

4 Result 2: the selected branch is not an attractor

Is the selected branch at least the time-asymptotic state? The URANS repeats say no. Figure 2b shows sliding-window means (0.1 s window) of the force coefficients against convective time $t^* = tU_\infty/c$ over a doubled horizon ($t^* = 0.81$–2.4,

Table 1: Branch-map proof: the refinement family at three incidences (NACA 0012, $Re = 10^6$, steady `simpleFoam`).

α	Grid	Cells	C_L	C_D	x_{sep}/c
5°	Medium	33,740	0.9663	0.0269	attached
5°	Fine	51,360	0.9724	0.0271	attached
15°	Coarse	16,100	1.1165	0.0694	0.53
15°	Medium	33,740	0.9915	0.0928	0.20
15°	Fine	51,360	1.2446	0.0605	attached
18°	Medium	33,740	1.0175	0.1333	0.17
18°	Fine	51,360	1.1417	0.1292	0.15

61 samples; both runs required a time-step-halved restart after transition-correlation failures, with continuous histories). Three facts stand out. First, the flow is violently unsteady: the drag standard deviation is 24% ($\alpha = 15°$) to 41% ($\alpha = 12°$) of the mean. Second, the windowed means *drift* — at neither angle do they settle toward the steady branch; the mean lift excess over steady grows to 27% and 34% (Table 2) and is still growing at the end of the window. Third, at $\alpha = 15°$ the steady and extended-window URANS drags *bracket* the measured value (the earlier short-horizon mean, 0.238, sat between the two): the horizon-averaged load of this non-stationary system is a function of the averaging window, so "the" URANS drag is as ill-defined as "the" steady drag.

Table 2: Non-attractor proof: steady branch vs. doubled-horizon URANS mean ± std ($t^* = 0.81$–2.4, $n = 61$) vs. Reynolds-matched experiment [2].

	C_L			C_D		
α	steady	URANS	exp.	steady	URANS	exp.
12°	1.039	1.322 ± 0.068	0.483	0.040	0.134 ± 0.055	0.022
15°	1.058	1.415 ± 0.108	0.331	0.075	0.307 ± 0.073	0.190

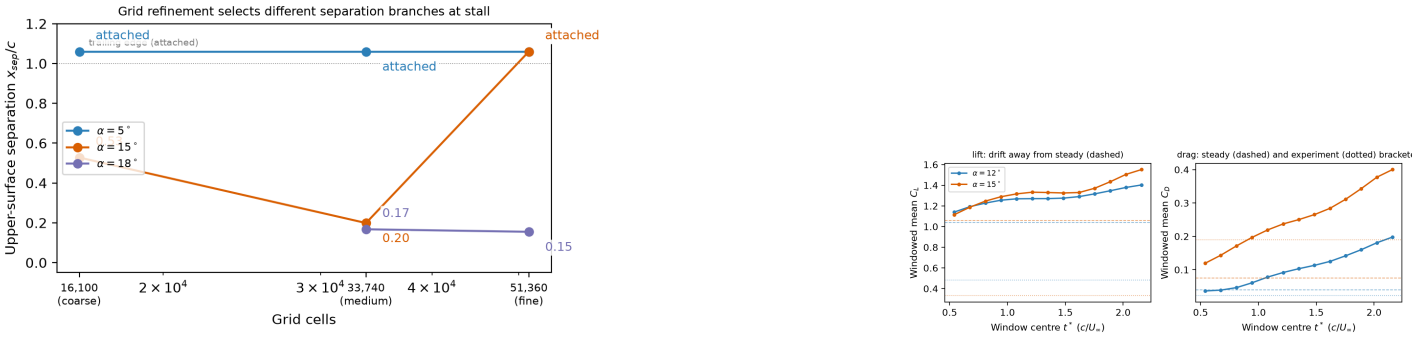


Figure 2: (a) Separation location through the refinement ladder: at stall onset ($\alpha = 15°$) each grid selects a different branch. (b) Sliding-window means over the doubled URANS horizon (dashed: steady; dotted: experiment [2]); at $\alpha = 15°$ (drag, right) steady and URANS bracket the measurement.

5 What follows for practice

The failure is stall-specific, and that specificity is the useful part: attached-flow certification remains exactly as trustworthy as the protocol claims. At stall onset we suggest three reporting rules: (i) run the refinement family *at the stall angle* and report the branch map — a single-grid stall result is a draw from a discrete set, not an estimate; (ii) pair the steady solution with a time-accurate repeat, since the two can bracket the experiment from opposite sides; (iii) state the averaging horizon in convective units. None of this costs more than the standard protocol. The deeper limitation remains three-dimensional [3, 4]: the branch structure documented here is a property of the model, not of the airfoil.

6 Conclusion

At trailing-edge stall onset, steady 2D RANS fails in a specific, diagnosable way: refinement hops between separation branches, iteration cannot choose among them, and the chosen branch repels the time-accurate dynamics. "Converged" at stall is a statement about a branch and a window, not about a number. All 125 cases, scripts, and diagnostics are openly released.

References

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